Programming for Business (CL-2016)

LAB INSTRUCTOR
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Week 04 Objectives:

- Conditional Statements
  - If
  - else
  - elif
  - Nested
  - Logical operations
  - Ternary Conditional Statement
  - Match Case

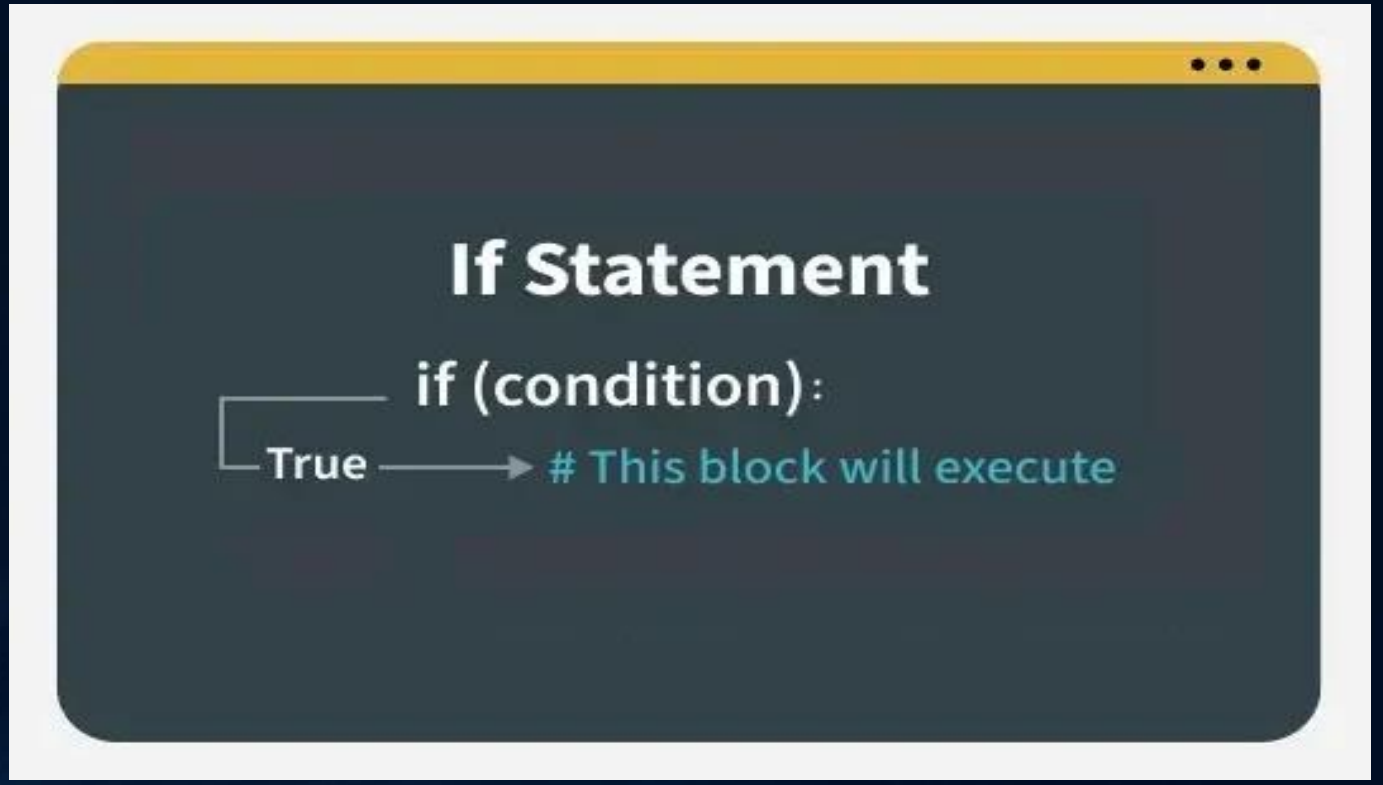
Conditional Statements in Python

- Conditional statements in Python are used to execute certain blocks of code based on specific conditions.

- These statements help control the flow of a program, making it behave differently in different situations.

If Conditional Statement

- If statement is the simplest form of a conditional statement. It executes a block of code if the given condition is true.



Example

- age = 20

  if age >= 18:

      print("Eligible to vote.")

- number = 15
  if number > 0:
      print("The number is positive")

Multiple Statements in If Block

- You can have multiple statements inside an if block. All statements must be indented at the same level.

Example

- Multiple statements in an if block:

```
age = 20
if age >= 18:
    print("You are an adult")
    print("You can vote")
    print("You have full legal rights")
```

Using Variables in Conditions

- Boolean variables can be used directly in if statements without comparison operators.
- Example
- Using a boolean variable:
- ```
  is_logged_in = True
  if is_logged_in:
      print("Welcome back!")
  ```
  ```

Short Hand if

- Short-hand if statement allows us to write a single-line if statement.

Example

```
age = 19

if age > 18: print("Eligible to Vote.")
```

- This is a compact way to write an if statement. It executes print statement if the condition is true.

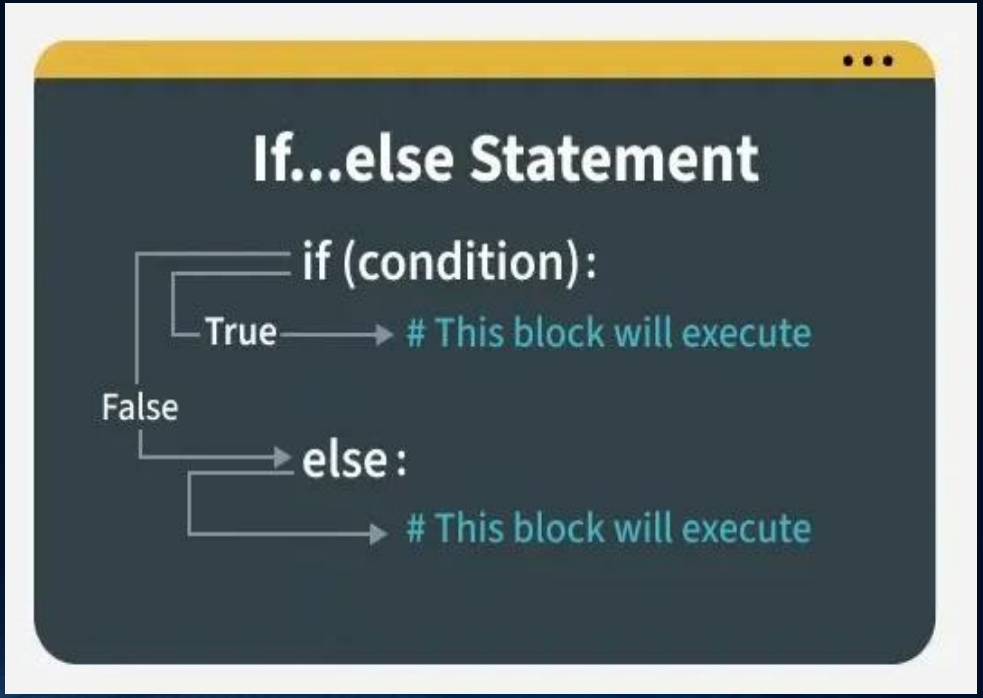
Indentation

- Python relies on indentation (whitespace at the beginning of a line) to define scope in the code. Other programming languages often use curly-brackets for this purpose.
- Example
- If statement, without indentation (will raise an error):

```
a = 33
b = 200
if b > a:
print("b is greater than a")
```

If else Conditional Statement

- If Else allows us to specify a block of code that will execute if the condition(s) associated with an if or elif statement evaluates to False.
- Else block provides a way to handle all other cases that don't meet the specified conditions.



Example

```
age = 10

if age <= 12:
    print("Travel for free.")
else:
    print("Pay for ticket.")
```

Example

- a = 200
- b = 33
- if b > a:
- print("b is greater than a")
- else:
- print("b is not greater than a")

Example

Validating user input:

- username = “Ali"

- if len(username) > 0:
- print(f"Welcome, {username}!")
- else:
- print("Error: Username cannot be empty")

Short Hand if-else

- The short-hand if-else statement allows us to write a single-line if-else statement.

```
marks = 45

res = "Pass" if marks >= 40 else "Fail"

print(f"Result: {res}")
```

Multiple Conditions on One Line

- You can chain conditional expressions, but keep it short so it stays readable:

Example

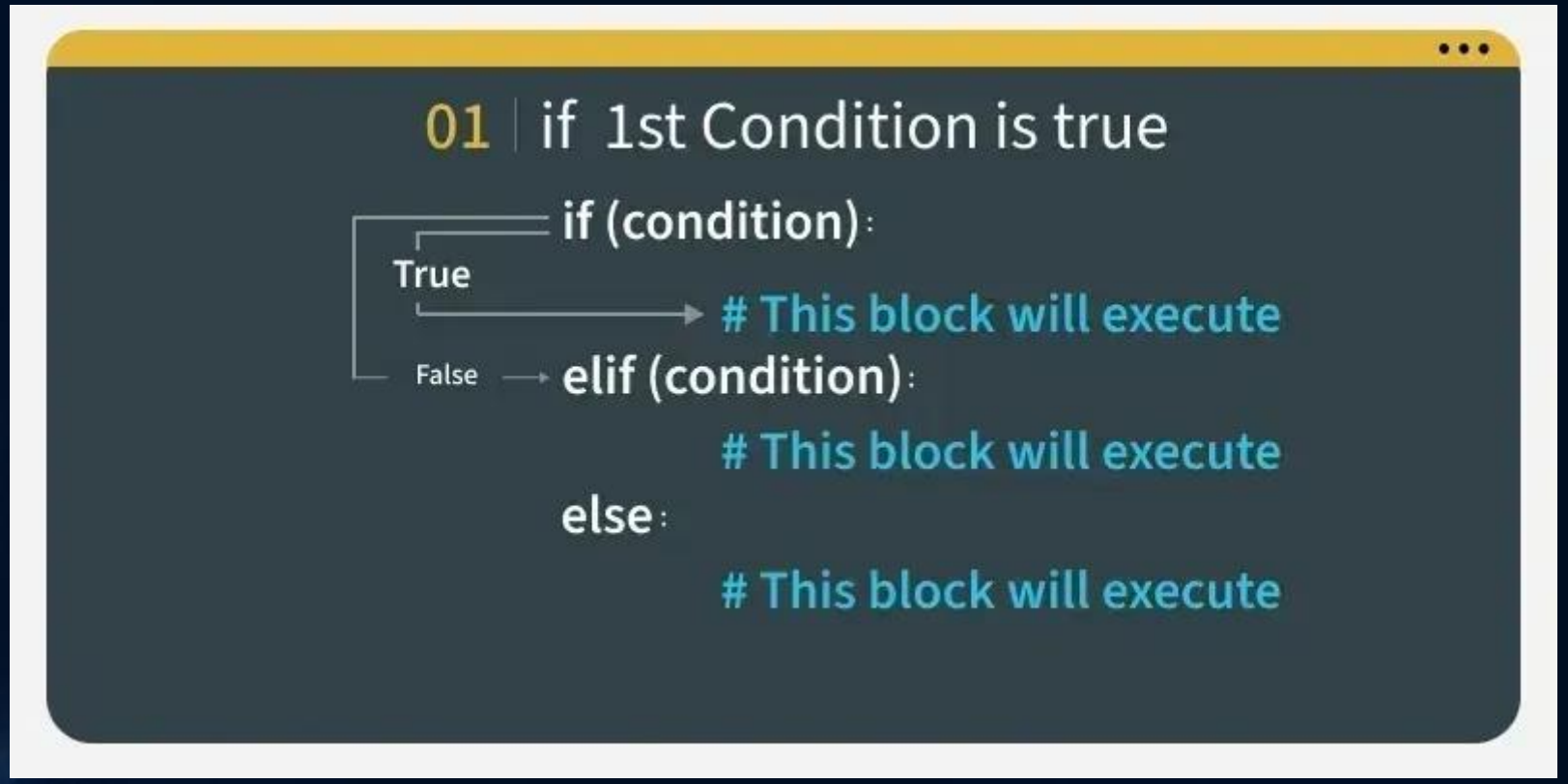
- One line, three outcomes:

```
a = 330
b = 330
print("A") if a > b else print("=") if a == b else print("B")
```

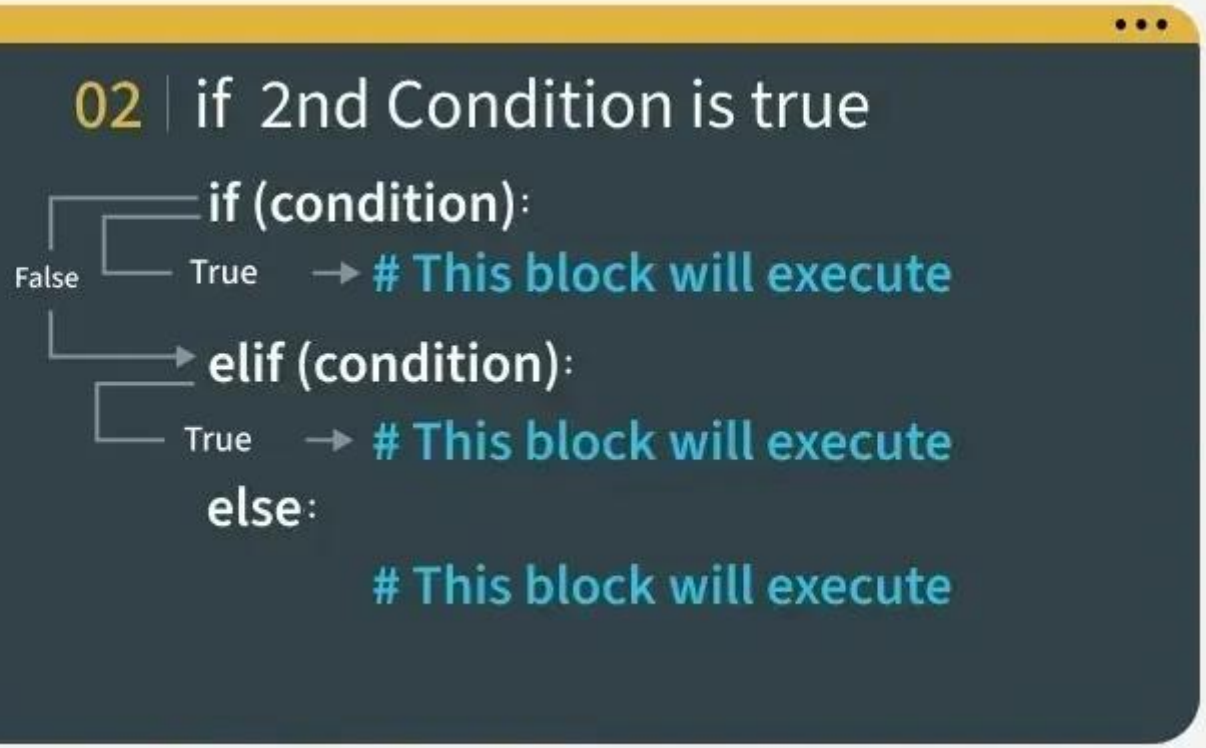
elif Statement

- elif statement in Python stands for "else if." It allows us to check multiple conditions, providing a way to execute different blocks of code based on which condition is true.

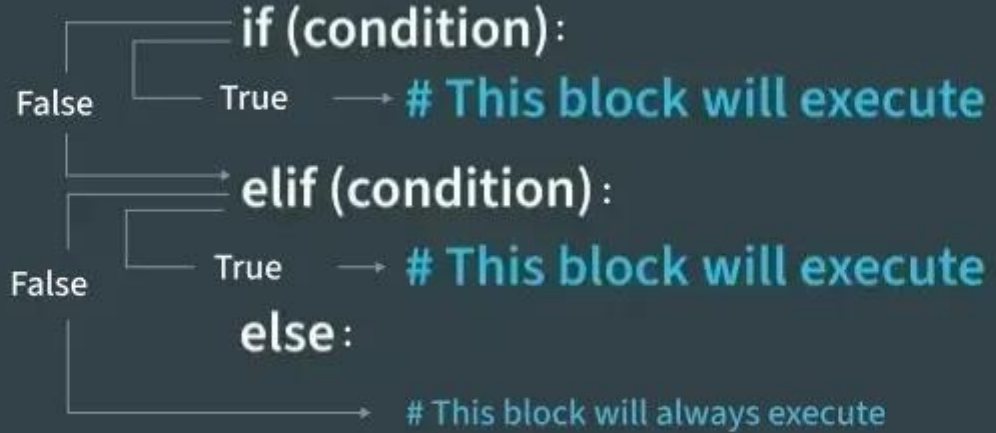
- Using elif statements makes our code more readable and efficient by eliminating the need for multiple nested if statements.


01 | if 1st Condition is true
if (condition):
True
This block will execute
False
elif (condition):
# This block will execute
else:
# This block will execute

02 | if 2nd Condition is true



03 | if all Condition are false



Example

```
age = 25
if age <= 12:
    print("Child.")
elif age <= 19:
    print("Teenager.")
elif age <= 35:
    print("Young adult.")
else:
    print("Adult.")
```

Testing multiple conditions:

- score = 75

```
if score >= 90:
    print("Grade: A")
elif score >= 80:
    print("Grade: B")
elif score >= 70:
    print("Grade: C")
elif score >= 60:
   print("Grade: D")
```

Example
Day of the week checker:

- day = 3

```
if day == 1:
  print("Monday")
elif day == 2:
  print("Tuesday")
elif day == 3:
  print("Wednesday")
elif day == 4:
  print("Thursday")
elif day == 5:
  print("Friday")
elif day == 6:
  print("Saturday")
elif day == 7:
  print("Sunday")
```

Example
Temperature classifier:

- temperature = 22

```
if temperature > 30:
  print("It's hot outside!")
elif temperature > 20:
  print("It's warm outside")
elif temperature > 10:
  print("It's cool outside")
else:
  print("It's cold outside!")
```

Python Logical Operators

Example

- Test if a is greater than b, AND if c is greater than a:

- a = 200
- b = 33
- c = 500
- if a > b and c > a:
- print("Both conditions are True")

Example

- Test if a is greater than b, OR if a is greater than c:

- a = 200
- b = 33
- c = 500
- if a > b or a > c:
- print("At least one of the conditions is True")

Example

- Test if a is NOT greater than b:

- a = 33
- b = 200
- if not a > b:
- print("a is NOT greater than b")

Combining Multiple Operators

- You can combine multiple logical operators in a single expression. Python evaluates not first, then and, then or.
- Combining and, or, and not:
- Example

```
age = 25

is_student = False

has_discount_code = True

if (age < 18 or age > 65) and not is_student or has_discount_code:

    print("Discount applies!")
```

Using Parentheses for Clarity

- When combining multiple logical operators, use parentheses to make your intentions clear and control the order of evaluation.
- Example
- Using parentheses for complex conditions:
- temperature = 25
  is_raining = False
  is_weekend = True

  if (temperature > 20 and not is_raining) or is_weekend:
    print("Great day for outdoor activities!")

Example

- User authentication check:
- username = “fastkhi"
  password = "secret123"
  is_verified = True

```
if username and password and is_verified:
  print("Login successful")
else:
  print("Login failed")
```

Example

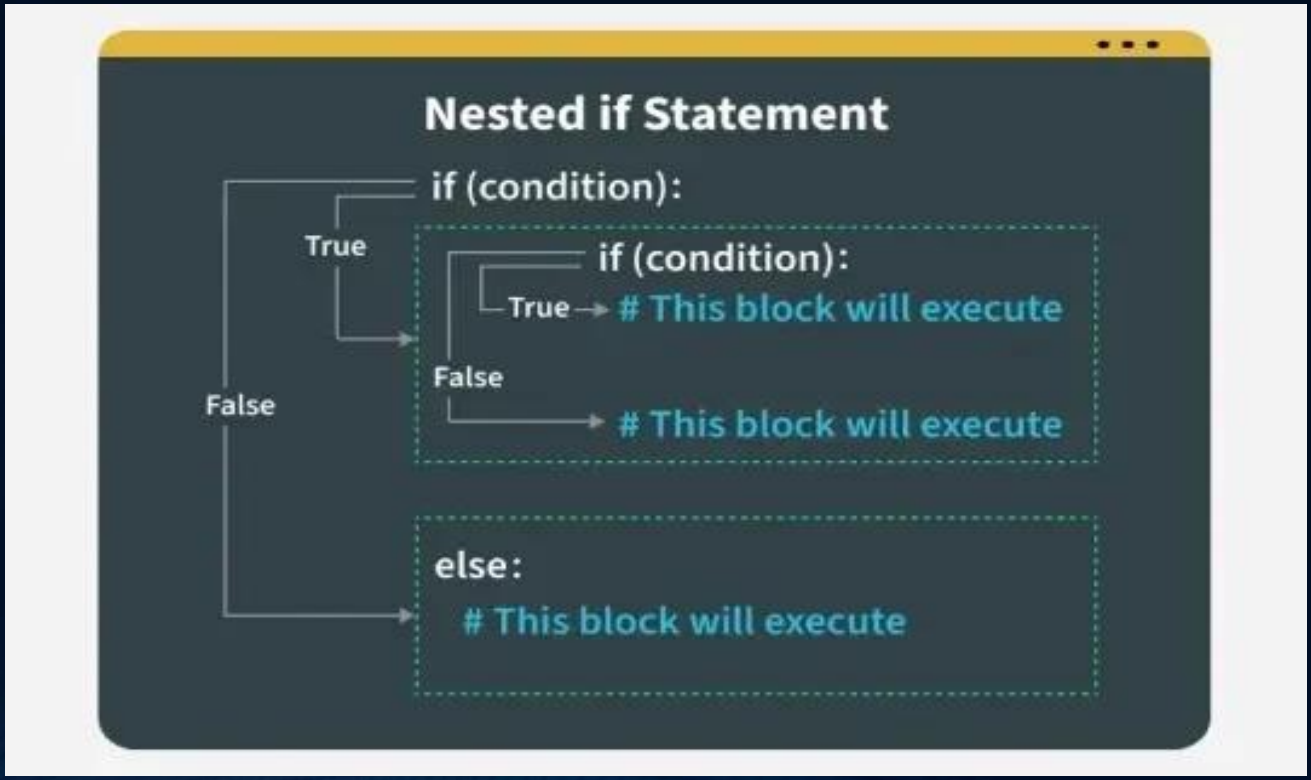
- Range checking with logical operators:

```
score = 85

if score >= 0 and score <= 100:
  print("Valid score")
else:
  print("Invalid score")
```

Nested if else Conditional Statement

- Nested if else means an if-else statement inside another if statement. We can use nested if statements to check conditions within conditions.



Example

```
age = 70
is_member = True


if age >= 60:
    if is_member:
        print("30% senior discount!")
    else:
        print("20% senior discount.")
else:
    print("Not eligible for a senior discount.")
```

Example

Three levels of nesting:

```
score = 85
attendance = 90
submitted = True

if score >= 60:
  if attendance >= 80:
    if submitted:
      print("Pass with good standing")
    else:
      print("Pass but missing assignment")
  else:
    print("Pass but low attendance")
else:
  print("Fail")
```

Ternary Conditional Statement

- A ternary conditional statement is a compact way to write an if-else condition in a single line. It’s sometimes called a "conditional expression."

```
# Assign a value based on a condition

age = 20

s = "Adult" if age >= 18 else "Minor"

print(s)
```

Match-Case Statement

- Match-case statement is Python's version of a switch-case found in other languages. It allows us to match a variable's value against a set of patterns.

```
number = 2

match number:
    case 1:
        print("One")
    case 2 | 3:
        print("Two or Three")
    case _:
        print("Other number")
```

Combine Values

- Use the pipe character | as an or operator in the case evaluation to check for more than one value match in one case:

Example

- day = 4
- match day:
- case 1 | 2 | 3 | 4 | 5:
- print("Today is a weekday")
- case 6 | 7:
- print("I love weekends!")

If Statements as Guards

You can add `if` statements in the case evaluation as an extra condition-check:

Example

```
month = 5
day = 4
match day:
  case 1 | 2 | 3 | 4 | 5 if month == 4:
    print("A weekday in April")
  case 1 | 2 | 3 | 4 | 5 if month == 5:
    print("A weekday in May")
  case _:
    print("No match")
```

Final Lab Task

- Ask the user to enter a transaction ID number.
  Print whether it is **Even** or **Odd**.
- If account balance is less than 500, show “Low Balance Warning”, otherwise show “Balance is sufficient”.
- Transaction amount > 100,000 AND Location is not "Pakistan"→ Show “Fraud Alert!”
- If payment method = "card" and amount > 5000 → 10% discount If method = "wallet" → 5% discount Otherwise → No discount
- Write a program to check if a year entered by the user is a leap year.